Constructors

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;

    Student () {
        name = "Aditya";
        age = 28;
        rollNo = 101;
        college = "IIT G";
    }
}
```

```
Student s1 = new Student();
```

Rules of Constructor

1. Same name as class.
2. No return type.
   Not even void.
3. Automatically called during Object Creation.
4. Used to initialize an Object
5. It can also be Overloaded.

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;

    Student () {
        //empty
    }
}
```

```
Student s1 = new Student();
```

Constructor is Mandatory.

⇓

Default Constructor.

Parameterized Constructor

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;

    Student (String n, int a, int rN, int c) {
        name = n;
        age = a;
        rollNo = rn;
        college = c;
    }

    Student () {
    }
}
```

```
Student s1 = new
        Student ("Aditya", 28, 101,
                "IITG");
```

```
Student s2 = new
        Student ();
```

this Keyword

```
Class Student {
    String name;
    int age;
    int rollNo;
    String college;
    Student (String name, int age, int rollNo, String College) {
        this.name = name
        this.age = age;
        this.rollNO = rollNo;
        this.College = "IITG";
}
```

Chaining in Constructor

↓

$\boxed{\text{Constructor Overload.}}$

↓

Chaining

Constructor → manually.

↳ X (NO)

Student s1 = new Student();

this(name, age);

Student(———); X

Run-time exception →

Heap

Not enough space in heap.